TEXORA · SESSIONAL NOTES · STUDY SHEET

Practical Study of the Working Aids

Fitting and sewing with a compensating foot, a hem folder and a binder, and proving their value the honest way: measured means and ranges against hand-guided work, with the time arithmetic to match.

TE-3108 · APPAREL MANUFACTURING TECHNOLOGY-II SESSIONAL · STUDY 10 · LEVEL 3 TERM 1

1. Aim and place of the study

Aim: to fit, set and sew with the principal working aids of TE-3107 Topic 5 — an edge-guide (compensating) foot, a double-turn hem folder and a binder attachment — measuring the consistency each aid actually delivers against hand-guided work.

2. Materials, machine settings and specimens

Materials and machines: SNLS machine; compensating feet of 2 mm and 6 mm offset; double-turn hem folder (30 mm input, 10 mm finished hem); raw-edge binder (24 mm strip input); poplin strips 40 × 8 cm (ten per aid), binding strips, ruler and seam gauge. **Method:** for each aid, sew five strips by the aid and five hand-guided, then measure the result at five points per strip — the study's conclusion is drawn from your own numbers.

Trouser measurement chart used throughout TE-3108 (men's basic trouser, garment values in cm): waist S 77 · **M 82** · L 87 · XL 92 (grade 5); hip 97 · **102** · 107 · 112 (grade 5); outseam (incl. waistband) 102 · **104** · 106 · 107; inseam 76 · **78** · 80 · 81; thigh 60 · **62** · 64 · 66 (grade 2); knee 44 · **46** · 48 · 50 (grade 2); hem (bottom) width 38 · **40** · 42 · 44 (grade 2); front rise 25 · **26** · 27 · 28. Size M is the working size unless the study states otherwise. Seam allowances, as in TE-2207/TE-2208: **1 cm** general, **1.5 cm** at side seams, waistband and other adjustment seams, **2.5 cm** at hems. The shirt values of TE-2207/2208 (size M chest 107, length 73, shoulder 46, sleeve 61.5, collar 40; grades 5/2/2/1.5/2 cm) are used unchanged wherever a shirt is the specimen.

3. Fitting and setting each aid

AID	FITTING/SETTING POINTS	SEW AND MEASURE

Compensating foot (6 mm)	Foot changed with the machine off; guide toe must ride the fabric edge squarely; needle position checked by hand-turning one revolution before power sewing.	Topstitch lines at a nominal 6 mm from the edge. Measure the distance at 5 points per strip.
Hem folder (double-turn)	Folder mounted so its mouth delivers the folded edge exactly under the needle line; scroll clean and polished; first strip fed slowly by hand-crank to prove the fold path.	Finished hem width (nominal 10 mm) at 5 points per strip; check the raw edge is fully enclosed.
Binder	Strip threaded through the scroll before mounting; binder centred so the edge enters mid-fold; needle must pierce both folds equally (Study 1 fault check).	Binding width face and back at 5 points per strip; the two faces must agree within 0.1 cm.

4. Recording the result

Tabulate, for each method (aid vs. hand), the five-strip means and the **range (max − min)** across all 25 measurements. The expected finding, which your table should confirm or honestly contradict: the aids' means sit on the nominal value and their ranges are a fraction of the hand-guided ranges — the quantitative form of TE-3107 Topic 5's claim that quality is set by steel. Record also the time per strip for both methods: the folder's time saving, multiplied out, is the worked economics of the theory sheet done with your own stopwatch.

Observation points — what to record

- Mean and range of every dimension, aid vs. hand — the core record of the study.
- Fold/binding faults and their setting cause (mouth alignment, strip threading, toe pressure).
- Time per strip for the hem, both methods; daily saving computed at 1,200 pieces.
- Any aid you could not make work, and the exact point at which it failed — an honest failure record passes; an invented success does not.

Precautions

- All foot and attachment changes with the machine switched off; hand-crank one full cycle after every change.
- Never force fabric through a folder: a folder that needs force is misaligned, and force hides the misalignment until the bundle is sewn.
- Measure with the gauge, not by eye — this study exists to replace impressions with numbers.

Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Your hem-folder results: mean 10.0 mm, range 0.6 mm; hand-guided: mean 10.4 mm, range 2.8 mm. State the two conclusions the examiner expects.
2. Why must the needle pierce both folds of a binding equally, and which earlier study's fault reappears if it does not?
3. A compensating foot's stitch line wanders although the toe rides the edge. Give two causes.

Answer key

1. Accuracy: the folder's mean is on nominal (10.0) while hand work drifts (10.4). Consistency: the folder's range is far smaller — the aid's real product is uniformity, with accuracy set at fitting.
2. Unequal piercing leaves one fold uncaught (it escapes in washing/wear) or the other over-caught and bulky. It is Study 1 sample C's fault: binding width unequal face-to-back, caused at feeding/setting.
3. The edge being followed is itself wavy or trimmed unevenly (the foot faithfully follows a bad edge), or the toe spring/pressure is weak so the foot rides up over thickness changes instead of tracking the edge.

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